ASSIGNMENT ONE

1. In a statistical study, what is the difference between an individual and a variable?

2. Are data at the nominal level of measurement quantitative or qualitative?

3. What is the difference between a parameter and a statistic?

4. For a set population, does a parameter ever change? If there are three different samples of the same size from a given population, is it possible to obtain three distinct values for the same statistic?

5. Numbers are often assigned to categorical data.

(a) Consider these number assignments for category items describing electronic ways of expressing personal opinions:

1 = Twitter; 2 = e-mail; 3 = text message; 4 = Facebook; 5 = blog
Are these numerical assignments at the ordinal data level or higher? Explain.

(b) Consider these number assignments for category items describing the usefulness of customer service:

1 = not helpful; 2 = somewhat helpful; 3 = very helpful; 4 = extremely helpful
Are these numerical assignments at the ordinal data level? Explain.

6. Lucy conducted a survey asking some of her friends to specify their favourite type of TV entertainment from the following list of choices:

Sitcom; reality; documentary; drama; cartoon; other

Do Lucy's observations apply to all adults? Explain. From the description of the survey group, can we draw any conclusions regarding the age of participants, the gender of participants, or the education level of participants? Explain

7. A national survey asked 1261 U.S. adult fast-food customers which meal (breakfast, lunch, dinner, snack) they ordered.
(a) Identify the variable.
(b) Is the variable quantitative or qualitative?
(c) What is the implied population?

8. Government agencies carefully monitor water quality and its effect on wetlands (Reference: Environmental Protection Agency Wetland Report EPA 832-R-93-005). Of particular concern is the concentration of nitrogen in water draining from fertilized lands. Too much nitrogen can kill fish and wildlife. Twenty-eight samples of water were taken at random from a lake. The nitrogen concentration (milligrams of nitrogen per litre of water) was determined for each sample.
(a) Identify the variable.
(b) Is the variable quantitative or qualitative?
(c) What is the implied population?

9. The archaeological site of Tara is more than 4000 years old. Tradition states that Tara was the seat of the High Kings of Ireland. Because of its archaeological importance, Tara has received extensive study (Reference: Tara: An Archaeological Survey by Conor Newman,

Royal Irish Academy, Dublin). Suppose an archaeologist wants to estimate the density of ferromagnetic artifacts in the Tara region. For this purpose, a random sample of 55 plots, each of size 100 square meters, is used. The number of ferromagnetic artifacts for each plot is determined.
(a) Identify the variable.
(b) Is the variable quantitative or qualitative?
(c) What is the implied population?

10. Categorize these measurements associated with student life according to level: nominal, ordinal, interval, or ratio.
(a) Length of time to complete an exam
(b) Time of first class
(c) Major field of study
(d) Course evaluation scale: poor, acceptable, good
(e) Score on last exam (based on 100 possible points)
(f) Age of student

11. Categorize these measurements associated with a robotics company according to level: nominal, ordinal, interval, or ratio.
(a) Salesperson's performance: below average, average, above average
(b) Price of the company's stock
(c) Names of new products
(d) Temperature (8°F) in the CEO's private office
(e) Gross income for each of the past 5 years
(f) Colour of product packaging

12. Categorize these measurements associated with fishing according to level: nominal, ordinal, interval, or ratio.
(a) Species of fish caught: perch, bass, pike, trout
(b) Cost of rod and reel
(c) Time of return home
(d) Guidebook rating of fishing area: poor, fair, good
(e) Number of fish caught
(f) Temperature of water

13. If you were going to apply statistical methods to analyse teacher evaluations, which question form, A or B, would be better?

Form A: In your own words, tell how this teacher compares with other teachers you have had.

Form B: Use the following scale to rank your teacher as compared with other teachers you have had.

1	2	3	4	5
worst	below average	average	above average	best

14. You are the manager of a specialty coffee shop and collect data throughout a full day regarding waiting time for customers from the time they enter the shop until the time they pick up their order. (a) What type of distribution do you think would be most desirable for the waiting times: skewed right, skewed left, or mound-shaped symmetric? Explain.

(b) What if the distribution for waiting times were bimodal? What might be some explanations?

15. A website rated 100 colleges and ranked the colleges from 1 to 100, with a rank of 1 being the best. Each college was ranked, and there were no ties. If the ranks were displayed in a histogram, what would be the shape of the histogram: skewed, uniform, or mound-shaped?